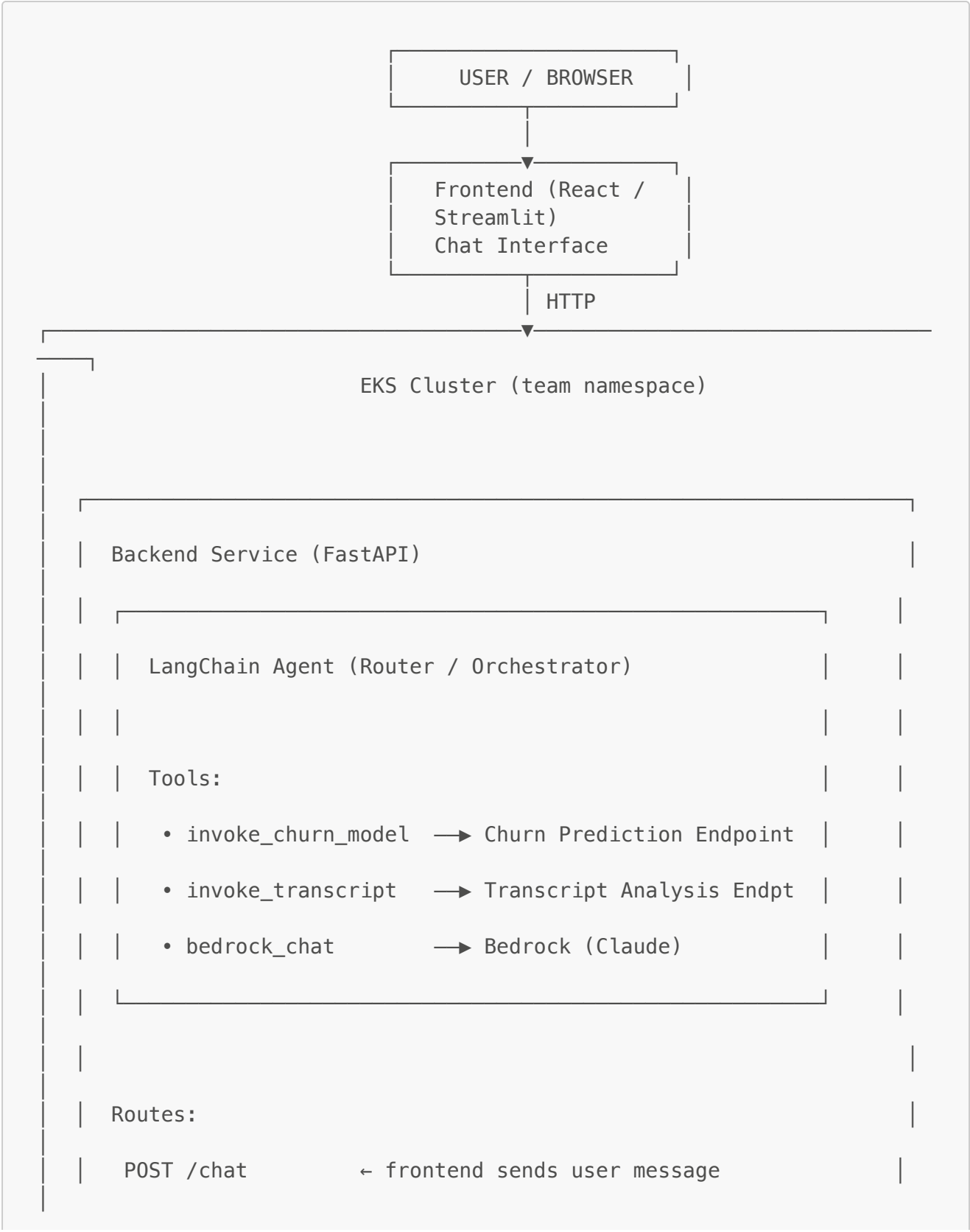
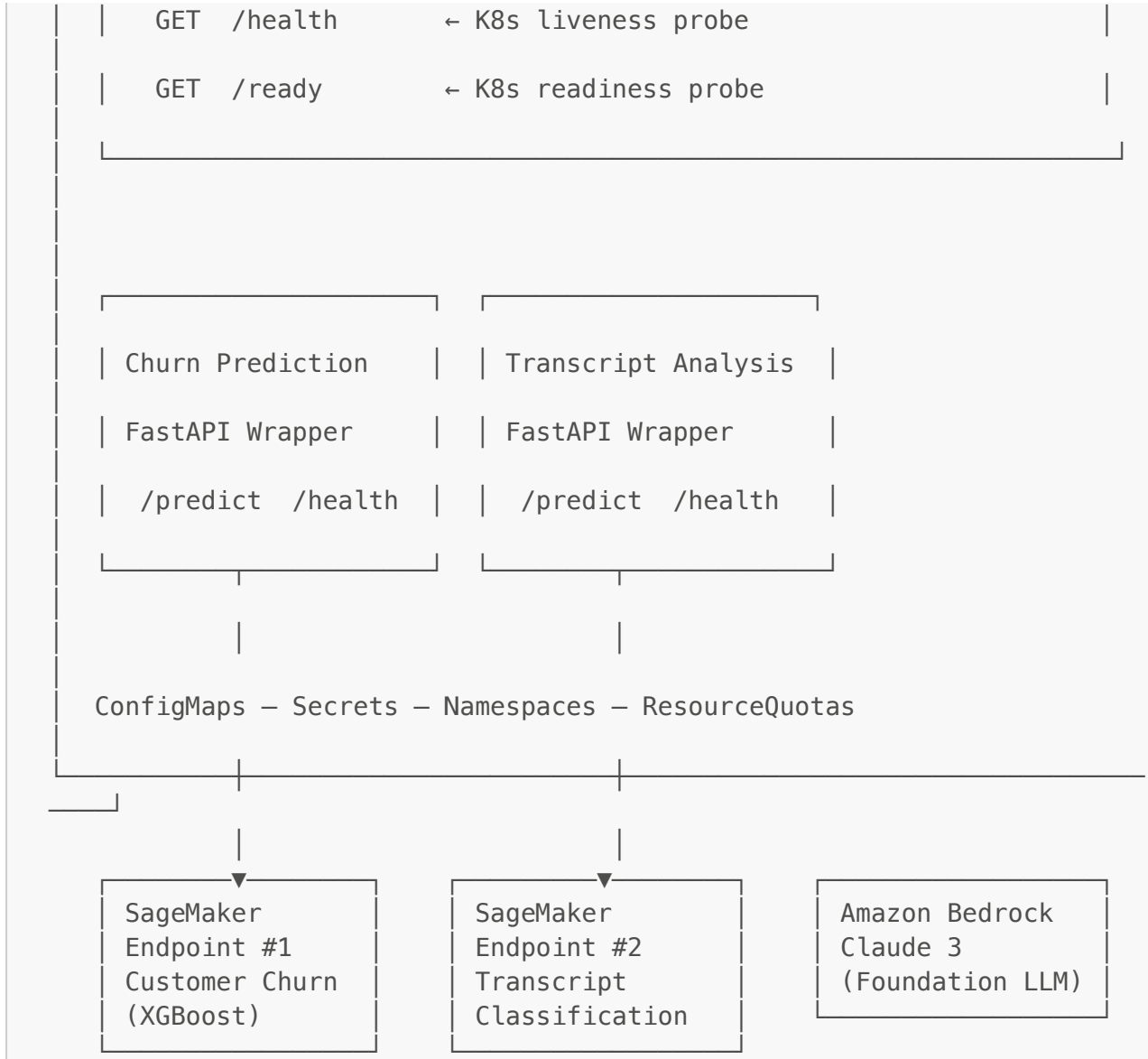
The Retention Engine — Project Roadmap

Team: Troy, Kathleen, Okino, George

Platform Architecture





INFRA: Terraform (S3, IAM, SageMaker, EKS references)
CI/CD: GitHub Actions → Build → Push GHCR → Deploy to EKS

Repo Structure

```
capstone/
├── .github/
│   └── workflows/
│       ├── terraform.yml            # Terraform plan/apply
(workflow_dispatch)
│       ├── sagemaker-deploy.yml     # Deploy/delete SageMaker
endpoints (workflow_dispatch)
│       ├── deploy.yml               # Docker build → GHCR → EKS
rollout (workflow_dispatch)
│       ├── ci-post-merge.yml        # Auto post-merge: tests → Docker
→ endpoints → E2E
│       └── slack-pr-events.yml      # Slack notifications on PR
```

```
open/merge
├── terraform/
│   ├── main.tf
│   ├── variables.tf
│   ├── outputs.tf
│   ├── providers.tf
│   └── modules/                    # (bonus) networking, iam, sagemaker
├── k8s/
│   ├── namespace.yaml
│   configmaps/
│      ├── agent-config.yaml
│   deployments/
│      ├── agent-deployment.yaml
│      ├── backend-deployment.yaml
│      ├── churn-wrapper-deployment.yaml
│      ├── frontend-deployment.yaml
│      ├── transcript-wrapper-deployment.yaml
│   ├── secrets.yaml             # (template — real values via GH
Secrets)
│   services/
│      ├── backend-service.yaml
│      ├── churn-wrapper-service.yaml
│      ├── transcript-wrapper-service.yaml
│      └── frontend-service.yaml
├── backend/
│   ├── Dockerfile
│   ├── app/
│   │   ├── main.py                 # FastAPI app, /chat, /health,
/ready
│   │   ├── agent.py                # LangChain agent + tool definitions
│   │   ├── tools/
│   │   │   ├── churn_tool.py       # Calls churn-wrapper /predict
│   │   │   ├── transcript_tool.py  # Calls transcript-wrapper /predict
│   │   │   └── bedrock_tool.py     # Bedrock invoke for general chat
│   │   └── config.py               # Env var config
│   └── requirements.txt
├── ml-wrappers/
│   ├── churn/
│   │   ├── Dockerfile
│   │   ├── app.py                  # FastAPI /predict + /health
│   │   └── requirements.txt
│   └── transcript/
│       ├── Dockerfile
│       ├── app.py                  # FastAPI /predict + /health
│       └── requirements.txt
├── sagemaker/
│   ├── churn/
│   │   ├── train.py                # Training script or notebook
│   │   ├── deploy.py               # Endpoint creation script
│   │   └── data/                   # Sample data / data prep scripts
│   └── transcript/
│       ├── train.py
│       ├── deploy.py
```

```
│       └── data/
├── frontend/
│   ├── Dockerfile
│   ├── src/                          # React or Streamlit app
│   └── ...
├── docker-compose.yaml               # Local dev: all services together
├── docs/
│   ├── architecture.md
│   ├── setup.md
│   ├── deployment.md
│   ├── teardown.md
│   └── llm-usage/                    # Per-member LLM usage docs
├── PROJECT_ROADMAP.md
└── README.md
```

Owner Matrix

Component	Primary Owner	Support	Key Deliverables
GitHub Projects / Kanban	**Troy**	All	Board setup, task cards, sprint tracking
GitHub Actions CI/CD	**Troy**	George	All `.yaml` workflows, secrets config
Dataset identification	**Kathleen + Okino**	—	Churn dataset, transcript dataset
SageMaker training + deploy	**Kathleen + Okino**	—	`sagemaker/` training scripts, live endpoints
ML FastAPI wrappers	**Kathleen + Okino**	Troy (CI)	`ml-wrappers/` services with `/predict` + `/health`
Terraform infra	**George**	Troy (CI)	`terraform/` — IAM, S3, SageMaker resources
Kubernetes manifests	**George**	Troy (deploy wf)	`k8s/` — all manifests
LangChain agent + Bedrock	**George**	Kathleen/Okino (tool contracts)	`backend/app/agent.py`, tool definitions
Backend FastAPI service	**George**	All	`backend/` — routes, agent wiring
Frontend chat UI	**All**	—	`frontend/` — chat interface
Documentation + diagrams	**All**	—	`docs/`, `README.md`

Graded Areas → Owner Mapping

#	Graded Area (10% each)	Primary Owner	Supporting
1	Containers & Dockerfiles	George (backend, frontend), Kathleen/Okino (ML wrappers)	Troy (CI builds)
2	Terraform Provisioning	George	—
3	CI/CD with GitHub Actions	Troy	All contribute workflows
4	LangChain Agentic Harness	Kathleen (LangGraph + LangSmith), George (initial scaffold)	Okino (tool contracts)
5	Bedrock Chat Agent & Frontend	George (Bedrock/agent), All (frontend)	—
6	SageMaker ML Endpoints	Okino (Endpoint 1: QA/Transcript), Kathleen (Endpoint 2: Churn)	George (integration)
7	Kubernetes Orchestration	George	Troy (CI/CD deploy steps)
8	Team Collaboration & Agile	Troy (board setup), All (participation)	—
9	Presentation & Documentation	All	—
10	Leveraging LLMs as Dev Tools	All (individual docs)	—

Interface Contracts

Define these early so everyone can work in parallel.

Churn Wrapper → SageMaker Endpoint

```
POST /predict
Request:  { "features": [0.5, 1.0, 3, 45.2, ...] }
Response: { "churn_probability": 0.82, "prediction": "churn" }

GET /health
Response: { "status": "healthy", "endpoint": "churn-endpoint-v1" }
```

Transcript Wrapper → SageMaker Endpoint

```
POST /predict
Request:  { "text": "I've been waiting for 30 minutes and nobody..." }
Response: { "category": "complaint", "sentiment": "negative",
"confidence": 0.91 }

GET /health
Response: { "status": "healthy", "endpoint": "transcript-endpoint-v1" }
```

Backend /chat → Frontend

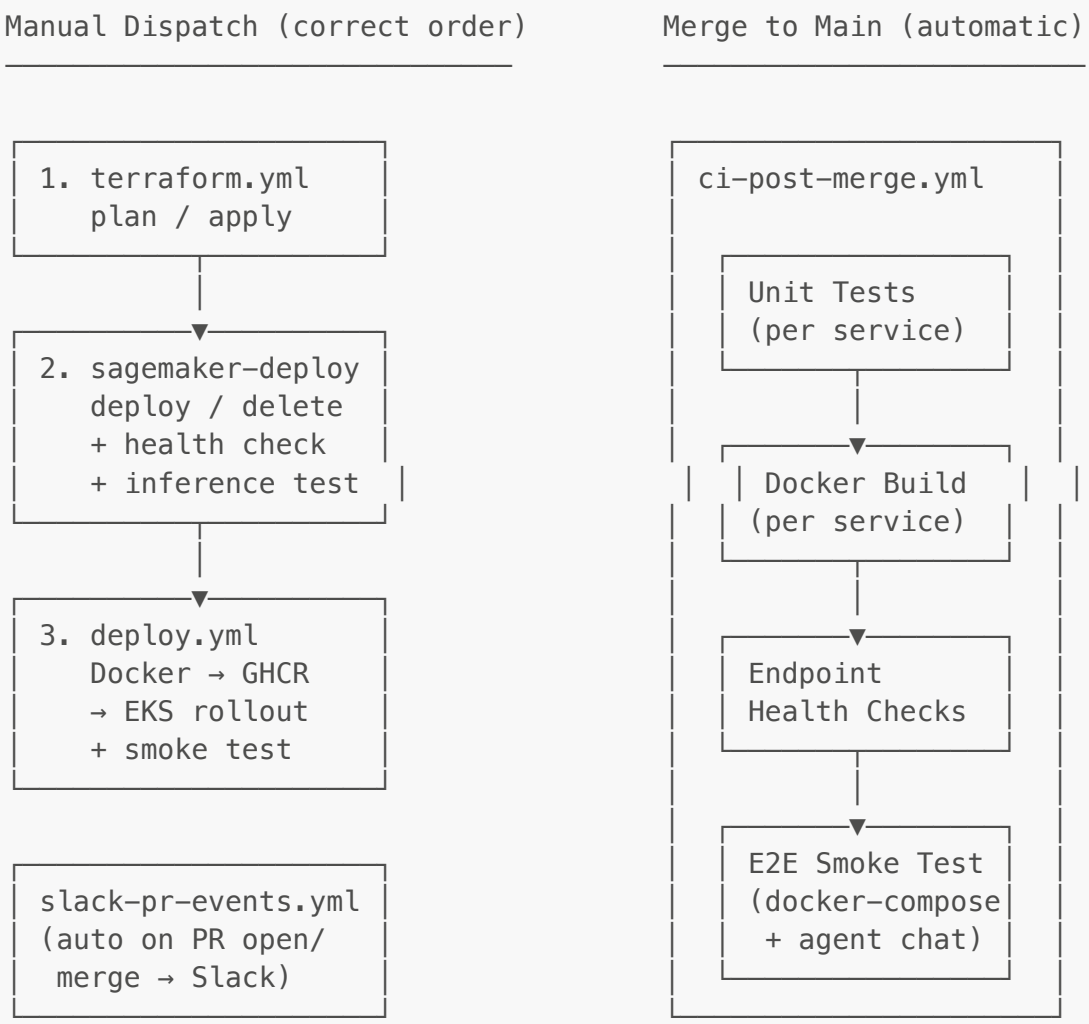
```
POST /chat
Request:  { "message": "What's the churn risk for customer 4521?" }
Response: { "response": "Based on the model prediction, customer 4521
has a 82% churn probability..." }

GET /health
Response: { "status": "healthy", "agent": "ready", "tools": ["churn",
"transcript", "bedrock"] }
```

LangChain Agent Design

```
# Simplified agent structure — George owns, Kathleen/Okino define tool
I/O

tools = [
    # Tool 1: Churn prediction
    #   Input: customer feature vector
    #   Action: POST to churn-wrapper-service:8000/predict
    #   Output: churn probability + label

    # Tool 2: Transcript analysis
    #   Input: raw text from service call
    #   Action: POST to transcript-wrapper-service:8000/predict
    #   Output: category + sentiment + confidence

    # Tool 3: Bedrock general chat
    #   Input: user message (when no tool needed)
    #   Action: bedrock invoke_model (Claude 3)
    #   Output: conversational response
]

# Agent routes based on user intent:
#   "Will customer X churn?"     → churn tool
#   "Analyze this transcript..." → transcript tool
#   "What can you help with?"    → bedrock general chat
#   Complex queries              → chain multiple tools
```

```
# Non-trivial agentic behavior:
#   If churn_probability > 0.7 → auto-generate retention offer via
Bedrock
#   Multi-step: transcript → QA score → combine with account data →
churn → retention offer
```

CI/CD Pipeline Overview



Workflows (Troy owns):

File	Trigger	What It Does
`terraform.yml`	`workflow_dispatch` (plan/apply, dev/staging/prod)	Terraform init → validate → plan → apply for S3, IAM, SageMaker
`sagemaker-deploy.yml`	`workflow_dispatch` (both/churn/sentiment, deploy/delete)	Deploy endpoints, poll for InService, validate inference

File	Trigger	What It Does
`deploy.yml`	`workflow_dispatch` (all/per-service, custom tag)	Matrix Docker build → GHCR push → kubectl apply → rollout restart → smoke test
`ci-post-merge.yml`	`push` to `main`	Unit tests → Docker build check → SageMaker health → E2E smoke test (docker-compose)
`slack-pr-events.yml`	PR `opened` / `closed`	Sends formatted Slack notification via webhook

Kubernetes Layout

```
Namespace: team-retention-engine
│
├── backend-deployment        (2 replicas)
│   └── backend-service       (ClusterIP or LoadBalancer)
│
├── churn-wrapper-deployment  (1 replica)
│   └── churn-wrapper-service (ClusterIP)
│
├── transcript-wrapper-deployment (1 replica)
│   └── transcript-wrapper-service (ClusterIP)
│
├── frontend-deployment       (1 replica)
│   └── frontend-service      (LoadBalancer → public)
│
├── configmap                 (endpoint URLs, region, model names)
└── secret                    (AWS creds — injected via GH Actions)
```

All inter-service calls stay **inside the cluster** (ClusterIP). Only the frontend service gets a LoadBalancer for public access. The backend talks to ML wrappers via K8s DNS (`churn-wrapper-service.team-retention-engine.svc.cluster.local:8000`).

Phase Overview

Presentation Date: April 23, 2026

```
Phase 0: Setup & Scaffolding        ██                    Mar 31 – Apr 1
Phase 1: Core Services              ████                  Apr 2 – Apr 8
Phase 2: Infra, Deploy & CI/CD      ██████                Apr 9 – Apr 15
Phase 3: Frontend, Polish & Docs    ████████              Apr 16 – Apr 22
Presentation Day                    ██████████            Apr 23
```

Phase 0 — Setup & Scaffolding (Mar 31 – Apr 1) ✅ COMPLETE

- [x] Kanban board, branch protection, workflow stubs, docker-compose
- [x] Terraform, K8s manifests, agent service scaffold, Dockerfiles, Bedrock access
- [x] TriLink transcript dataset, sentiment-analysis-api scaffold, training notebook
- [x] TriLink churn dataset, sagemaker/churn scaffold, churn-predictor-api scaffold
- [x] API contracts agreed, React + Vite + Tailwind chosen, LLM usage logging started

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Phase 1 — Core Services (Apr 2 – Apr 8) ✅ MOSTLY COMPLETE

Troy — CI/CD Pipelines ✅

- [x] All 5 workflow files implemented (terraform, sagemaker-deploy, deploy, ci-post-merge, slack-pr-events)
- [x] Manual dispatch for infra/deploy with correct execution order
- [x] Automatic post-merge validation with unit tests, Docker builds, endpoint health, E2E

Okino — QA/Transcript Evaluator

- [x] Training notebook (`services/sentiment-analysis-api/sentiment_training.ipynb`)
- [x] FastAPI wrapper with Bedrock-powered sentiment analysis
- [x] Dockerfile, K8s deployment + service manifests
- [x] Terraform `sagemaker.tf` for endpoint config
- [ ] SageMaker endpoint deployment — in progress
- [ ] `/predict` route must return: qa_score, sentiment, emotion_frustration, emotion_anger, sentiment_shift, escalation_flag, resolution_flag

Kathleen — Churn Predictor ✅

- [x] XGBoost model v3: 95% accuracy, 0.9861 AUC, 31 features
- [x] Cross-model integration: 7 Agent 1 features from synthetic call data
- [x] SageMaker endpoint deployed (XGBoost container, native format) — InService
- [x] FastAPI wrapper with internal customer data lookup from S3
- [x] `/predict` accepts customer_id + Agent 1 fields, looks up account data internally
- [x] `/customers` API for frontend searchable dropdown
- [x] `/customer-details/{id}` API for agent tool
- [x] `/high-risk` API with batch SageMaker prediction + caching
- [x] Dockerfile (multi-stage build, slim base, health check)

Kathleen & Okino — Orchestration ✅

- [x] retention_agent.py: 4-tool agent (get_customer_details, analyze_call, predict_churn, get_high_risk_customers)
- [x] TriLink product catalog with approved retention actions per risk level
- [x] Output guardrails validating agent recommendations
- [x] Conversation memory (InMemoryChatMessageHistory, session-based)

- [x] Intelligent routing: agent decides tools based on question type
- [x] churn_tool.py: sends customer_id + 7 Agent 1 fields
- [x] high_risk_tool.py: fetches ranked at-risk customer list
- [x] customer_tool.py: looks up account details for conversational queries

George — Infrastructure & Agent

- [x] Terraform: S3, IAM roles, sagemaker.tf, iam.tf, s3.tf
- [x] K8s: all deployments, services, configmaps, secrets, namespace quota
- [x] Agent service: app.py with /chat route, CORS, session_id support
- [x] docker-compose.yml for full local stack
- [ ] Deploy full stack to EKS

Kathleen — Frontend ✅

- [x] React + Vite + Tailwind Manager's Command Center
- [x] **Analyze tab**: searchable customer dropdown, RiskCard, SentimentCard, ActionCard
- [x] **Chat tab**: conversational Bedrock interface with smart fallback
- [x] Lucide React SVG icons (professional)
- [x] Dockerfile (multi-stage: node → nginx)

Phase 2 — Infrastructure, Deploy & CI/CD (Apr 9 – Apr 15)

Troy — Deploy Pipeline ✅

- [x] `terraform.yml` — plan/apply with environment selection (dev/staging/prod)
- [x] `sagemaker-deploy.yml` — deploy/delete endpoints with health polling + inference validation
- [x] `deploy.yml` — matrix build → GHCR → EKS rollout with smoke test
- [x] `ci-post-merge.yml` — 4-job automatic pipeline on push to main
- [x] `slack-pr-events.yml` — Slack notifications on PR open/merge

Okino — Finish Endpoint 1

- [ ] Deploy sentiment model to SageMaker endpoint
- [ ] Confirm `/predict` output matches churn_tool.py expected fields
- [ ] Test end-to-end: transcript → Agent 1 → Agent 2 → Agent 3

George — K8s Deploy

- [ ] Deploy full stack to EKS
- [ ] Verify Bedrock + SageMaker access from inside cluster
- [ ] Test full agent pipeline from deployed frontend

Kathleen & Okino — Integration Testing

- [ ] Test full 3-agent pipeline end-to-end with live endpoints
- [ ] Verify high-risk batch prediction works on EKS

Phase 3 — Frontend, Polish & Docs (Apr 16 – Apr 22)

Frontend — Manager's Command Center

- [x] Analyze tab with customer dropdown + results panel
- [x] Chat tab with Bedrock conversational interface
- [x] Lucide icons, Tailwind styling
- [ ] "High Risk" leaderboard dashboard panel (bonus)

Memory Enhancement (Bonus — if time allows)

- [ ] Integrate mem0 for persistent semantic memory (reference: CyberRisk portfolio project)
- [ ] Requires: PostgreSQL + pgvector (could use AWS RDS)
- [ ] Would enable: long-term user preferences, cross-session context, semantic search over past analyses
- [ ] Alternative: current InMemoryChatMessageHistory is sufficient for demo

Troy — Final CI/CD & Collaboration

- [ ] Wire frontend CI/CD
- [ ] Final Kanban cleanup
- [ ] Write docs: CI/CD setup guide, teardown steps

Kathleen & Okino — ML Docs & Bonus

- [x] Kathleen's SageMaker endpoint verified InService
- [x] Kathleen LLM usage documented (`docs/llm-usage/kathleen-llm-usage.md`)
- [ ] Okino's SageMaker endpoint — needs deployment
- [ ] Okino LLM usage doc
- [ ] Ensure model invocations work from inside K8s pods

George — Infrastructure Finalization

- [ ] Verify Terraform state is clean and reproducible
- [ ] Terraform remote state with S3/DynamoDB (bonus)
- [ ] Architecture diagram
- [ ] Write docs: setup, deployment, teardown steps

Bonus Integrations (Anyone)

- [ ] Amazon Transcribe pipeline: S3 audio → Transcribe → feed to agent
- [ ] Amazon Polly: text-to-speech for agent responses
- [ ] Model drift tracking — lightweight `/drift` endpoint in churn predictor:
  - Log prediction distributions over time (rolling window)
  - Compare incoming feature distributions to training baselines
  - Return drift metrics (mean shift, distribution divergence)
  - Alert when predictions skew beyond threshold
  - No SageMaker Model Monitor needed — runs inside the FastAPI wrapper

Documentation (All Members)

- [ ] **README.md** — project overview, team members, quick start
- [ ] **docs/setup.md** — full reproduction steps (clone → provision → deploy → test → teardown)
- [ ] **docs/architecture.md** — at least one architecture diagram:
  - System-level diagram showing all services and data flow
  - Deployment pipeline diagram
  - (Bonus) C4 diagrams, sequence diagrams
- [ ] **docs/llm-usage/** — each member documents:
  - Which LLM tools they used
  - Specific examples of code generation, debugging, architecture planning
  - Critical evaluation: what they accepted, rejected, or modified
  - (Bonus) comparison of LLM-generated vs hand-written code

Presentation Prep

- [ ] Assign presentation sections — each member covers their area:
  1. **Business Problem & Architecture Overview** — any member (2 min)
  2. **Infrastructure & Terraform** — George (3 min)
  3. **CI/CD & Git Workflow** — Troy (3 min)
  4. **ML Endpoints & Data Pipeline** — Kathleen & Okino (3 min)
  5. **Agentic Layer, Bedrock, K8s & Frontend** — George (3 min)
  6. **Live Demo** — all (3 min)
  7. **LLM Usage & Lessons Learned** — all (2 min)
- [ ] Rehearse at least once as a team
- [ ] Prepare for Q&A on any section

Final Checks

- [ ] GitHub Projects board is up to date with task history
- [ ] All PRs have descriptions and at least one review
- [ ] Commit history shows contributions from all members
- [ ] No AWS credentials in the repo (scan with `git log --all -p | grep -i "AKIA"`)
- [ ] All services healthy on EKS
- [ ] CI/CD pipelines green

Milestone ✓

> Demo-ready. Clone → follow docs → full deployment reproducible.

Parallel Work Strategy

The key to speed: **everyone works simultaneously from Day 1**.

```
                Troy            Kathleen/Okino        George
                ────            ──────────────        ──────
```

```
Week 1 Focus:       CI/CD          ML Models              Infra + Agent
                    Repo           Datasets               Terraform
                    Workflows      SageMaker Train        K8s Manifests
                                   FastAPI Wrappers       LangChain

Week 2 Focus:       Frontend       Frontend               Frontend
                    Final CI       Hardening               Deploy to EKS
                    Docs           Docs                    Docs
```

Integration points (where people need to sync):

1. **API contracts** — agree on request/response shapes before coding (Day 1)
2. **Docker image names** — Troy needs image paths for CI, George needs them for K8s
3. **Endpoint URLs** — Kathleen/Okino provide SageMaker endpoint names, George puts them in ConfigMaps
4. **Secrets** — Troy sets up GH Secrets, George references them in K8s Secrets

---

Bonus Points Checklist

Area	Bonus Item	Difficulty	Owner	Status
Containers	Alpine/slim bases, health checks, `.dockerignore`	Low	All	✅ Done
Terraform	Remote state with S3/DynamoDB	Low	George	Pending
Terraform	Modular structure (separate modules)	Medium	George	Pending
CI/CD	Separate CI workflows per service	Low	Troy	✅ Done — matrix builds in deploy.yml + ci-post-merge.yml
CI/CD	Branch targeting, rollback, multi-workflow chaining	Medium	Troy	✅ Done — workflow_dispatch ordering, post-merge auto-validation
LangChain	LangGraph workflows	Medium	Kathleen	✅ Done — retention_graph.py
LangChain	LangSmith observability	Medium	Kathleen	✅ Done — retention-engine project
Bedrock/Chat	Conversation memory persistence	Medium	Kathleen	✅ InMemory done, mem0 stretch
Bedrock/Chat	Polished UI (Tailwind/MUI)	Medium	Kathleen	✅ Tailwind + Lucide

Area	Bonus Item	Difficulty	Owner	Status
SageMaker	Third endpoint or model versioning	Medium	Kathleen/Okino	Pending
SageMaker	Retry logic, error handling, fallback	Medium	Kathleen	✅ Fallback in chat
K8s	ResourceQuota + LimitRange	Low	George	✅ Done
K8s	HPA, persistent storage, rolling updates	Medium	George	Pending
Collaboration	Sprint artifacts, retro notes, standup docs	Low	Troy + All	Pending
Collaboration	Code review comments on PRs	Low	All	✅ Done (40+ PRs)
Presentation	C4/sequence diagrams	Medium	All	Pending
Presentation	Present early	Low	All	Pending
LLM Usage	Compare LLM vs hand-written code	Medium	Kathleen	✅ In llm-usage doc
LLM Usage	Document prompt engineering techniques	Low	Kathleen	✅ In llm-usage doc
Extra	AWS Transcribe/Polly integration	Medium	Anyone	Pending
Extra	Blog articles, MkDocs, video series	High	Anyone	Pending
Extra	Portfolio page integration	High	Anyone	Pending

Daily Standup Template

```
**Name:**
**Yesterday:** What I completed
**Today:** What I'm working on
**Blockers:** Anything stopping progress
**LLM Note:** How I used LLM tools today (for documentation)
```

Key Principles

1. **Vertical slice first** — get one path working end-to-end before expanding

2. **Parallel lanes** — infrastructure and application work happen simultaneously
3. **API contracts on Day 1** — agree on request/response shapes so everyone can work independently
4. **Feature branches always** — never commit directly to `main`
5. **Document as you go** — don't leave it all for Phase 3
6. **Log your LLM usage** — it's 10% of the grade, treat it seriously
7. **Test locally before deploying** — Docker Compose and Minikube first, EKS second
8. **Reuse prior work** — FastAPI services, K8s manifests, Terraform configs from earlier modules are fair game